

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE:CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I Mukkaram (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mukkaram

15
20

(Signature)

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

03-08-2026

Artificial Intelligence

Faculty name: Dr. Teena

Class Test.

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AI & DS

1. Introduction:

An artificial intelligence agent which senses the environment using the sensors and takes actions using actuators like temperature detector, AI based VAV intelligence, vacuum cleaner, chatgpt, etc.

Types of Artificial Intelligence agent:

1. Simple reflex agent:

It operates on the current task given to it by the agent

2. Goal based agent:

Goal based agent moves to a specific end point or goal from its initial state for work completion.

3. Utility based agent:

It proceeds with a point at which the agent has highest utilization.

4. Model based reflex agent:

It works on the futuristic action and does the work.

Conclusion:

A hybrid agent is perfect for the smart home as model based reflex agent creates a virtual model for environment and adapts to it.

2. Introduction:

If we have an AI based robot in a Maze it does not get free on its first attempt, it chooses a path from its initial position and adapts to the surrounding different ways and creates its own map and chooses its path.

* Uniform cost search:

It calculates the path cost for every path and proceeds with the minimum path cost.

* Iterative dependency search:

It finds the depth of every path for its traversal. ~~It~~ It calculates the distance occurred

Therefore, if an AI based robot in a maze is placed, it does not accurately find the right way on its first try. It chooses a path, ~~for~~ after several ways by creating a map by itself and chooses the path using its sensors and actuators.